

Social Influence Models based on Starbucks Networks

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Abstract

Starbucks coffee shops have been spread rapidly and widely all over the world, which implies that there may be diffusive powers among them and thus can be represented as social networks. In particular, the spreading speed of Starbuck Korea was at record levels [10]. In this paper, we constructed social networks using the information about Starbuck Korea (ex. latitude and longitude of each Starbucks store in Korea, the opening date of them, opening orders of them, etc.) and evaluated influence scores of each store to measure the spreading power of Starbucks in Korea. Here, we proposed two network evaluation models, Dynamic Influence Model and Static Influence Model. Through these models, we can represent location based social networks and evaluate each node's diffusive power for expanding the size of networks and for spreading coverage all over the network.

1 Introduction

Starbucks Korea opened the 100th store only after 5 years in Korea, which implies that the stores have been spread most rapidly in the world [10]. In this paper, we consider Starbucks stores as an exemplary social network showing the word-of-mouth effect, because this record could be achieved without any TV commercials; new customers have been introduced to Starbucks by their friends who had visited it before, or they have visited to the nearest store by themselves after watching office workers with Starbucks coffees in their hands.

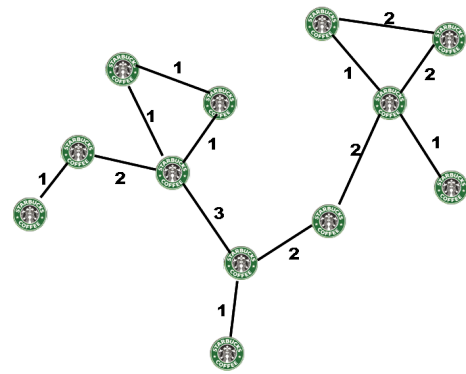


Figure 1. Starbucks Network as a Graph

We constructed a network by representing each store as a node and connecting them according to the distance between them as Fig. 1. Edge weight is decided by the radius around a particular store; if a store is in the specific radius around its neighbor as 200m, 400m, or 600m, then the value of weight is assigned as 1, 1/2, or 1/3, respectively. Here, we defined two kinds of influence models and analyzed the diffusive power of each node in a network using the Starbucks Korea network data from [2]. Through the models, we can tell why Starbucks stores in Korea have expanded so fast and which store played the most important role in the growth of the network.

The first model is the Dynamic Influence Model for time-varying diffusion analysis. As time goes by, the distribution of network varies, and the way of spreading innovations also changes. We name it *dynamic network*. This model estimates how much each node contributes to increasing the

scale of dynamic networks. With this model, we can trace the history of a social network formation and find which node was the most influential in diffusion of innovation.

The second model is the Static Influence Model for cumulative spreading power analysis. With this model, we can find which node is the most influential so at this time in a given network; it does not consider the temporal aspect of spreading.

We achieved results from the proposed models above and figured out the features of the most influential store in expanding the size of Starbucks network in Korea. We believe that these models can be applied for effective diffusion targeting in various fields, particularly, when considering location based social networks.

In section 2, we will examine the related works for evaluating diffusive power of networks and then, in section 3, we will explain how we collected data and how we chose the data for this research finally. In section 4, we will describe the proposed models and their basic common mechanisms, and in section 5, we will show the results from the models. Finally, we will conclude our research and discuss the limitations of our work.

2 Related Works

Diffusion of innovation [13][4][3] is gradually processed throughout the channels in social networks. Rogers [11] also mentioned that innovation is gradually spread through the communication channels of social members who are the parts of Social System. In other words, diffusion process consists of mainly four elements; ① Innovation, ② Communication Channels, ③ Time, ④ Social System [9]. Our research includes these constituents as well as follows. ‘① The settlement of coffee culture in Korea (Innovation)’ is fundamentally made by ‘② word of mouth (Communication Channel)’. The increased number of visitors of a Starbucks store affects the opening of a new shop near the store. As ‘③ time (Time)’ goes by, the number of Starbucks stores increases and finally ‘④ Starbucks store network system (Social System)’ is built. Furthermore, we think that the settlement of coffee culture in Korea is deeply related to the diffusive power of networks and we tried to analyze influence models based on these four constituents.

The topic of ‘Diffusion of Innovation’ has been studied in various fields such as social science, anthropology, mathematics, computer science, and so on [14][12]. Young considered each node in a social network as an agent, and concluded that each agent accepts innovations according to inherent factors (they are different from one another in in-nateness) and external factors (ex. the number of neighbors who accepted the innovation already) [15]. Goldenberg also studied the relationship between the diffusion processes and the structures of social networks. Finally, he found that

weak ties which have various kinds of relations to different kinds of groups played a more important role of diffusion rather than strong ties do [6]. In addition, probabilistic models have been studied for maximizing the diffusion coverage and targeting the optimal node as a seed node [5][8][7].

While the existing researches evaluate the influential power of each node based on the network coverage in a fixed size of network, this study evaluates an extent of contribution to diffusion in a dynamically changing network. In addition, we tried to analyze Starbucks chains as a social network from a new point of view. We expect this research can be applied to various kinds of area for estimating the nodal influential power of dynamic social networks.

3 Starbucks Data Analysis

The Starbucks Korea homepage provides with the information of store names, addresses, the opening dates, and so on. [2] We transformed store addresses into the information of latitude and longitude. Also, from the Coffee Bean and Tea Leaf Korea homepage, we obtained the names and addresses of stores for analyzing the relationships between two major coffee brands in Korea. However, this homepage does not offer the opening date of each store [1]. We also extracted the information of latitude and longitude of the Coffee Bean stores in the same way.

The number of Starbucks stores in Korea is as Fig. 2. 156 stores in Seoul, 34 stores in Gyeonggi Province, 16 stores in Busan, 8 stores in Daegu, 7 stores in Gangwon Province, 5 stores in Gyeongsang, 4 stores in Gwangju, 3 stores in Incheon, 3 stores in Chungcheong Province, 2 stores in Daejeon, 2 stores in Ulsan (May, 2008). In this research, we analyzed the 156 stores in Seoul, because the number of stores in other areas is relatively too small to construct a network.

In addition, Starbucks stores in Seoul are densely populated especially in Gangnam-gu (43 stores, 28%), Jung-gu (18 stores, 12%), Jongno-gu (16 stores, 10%), and Seocho-gu (15 stores, 10%); the maximum distance between any two stores is under 600m. As Fig. 3 shows, the rest of stores outside of those four districts are scattered. Therefore, it is not appropriate to analyze network influence flows among the outside stores.

Fig. 3. represents Starbucks networks in Seoul; nodes represent Starbucks stores and edges indicate that they are located closely (less than 600m) each other according to the information of latitude and longitude. Here, we can find that the stores in Seoul can be mainly divided into two clusters, (1) Jung-gu / Jongno-gu cluster (pink circles) and (2) Gangnam-gu / Seocho-gu cluster (blue circles); Jung-gu and Jonno-gu adjoin each other and also Gangnam-gu is adjacent to Secho-gu. Therefore, in this research, we focused on the 92 Starbucks stores located in the main districts

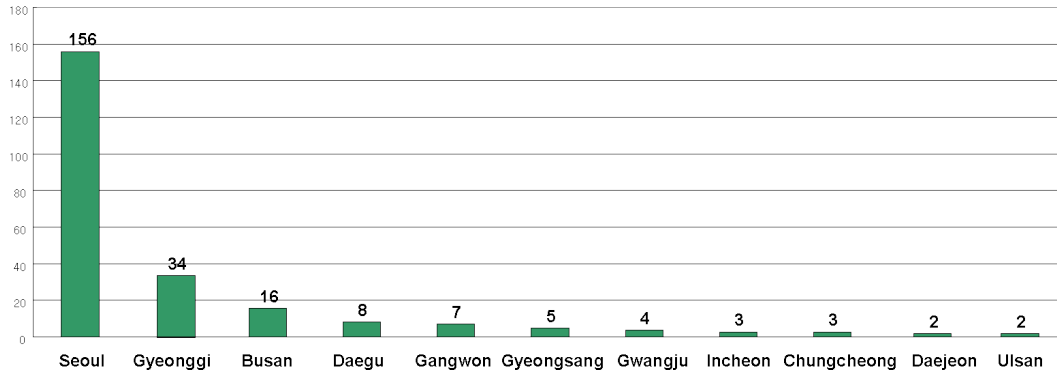


Figure 2. The Number of Starbucks stores in Korea

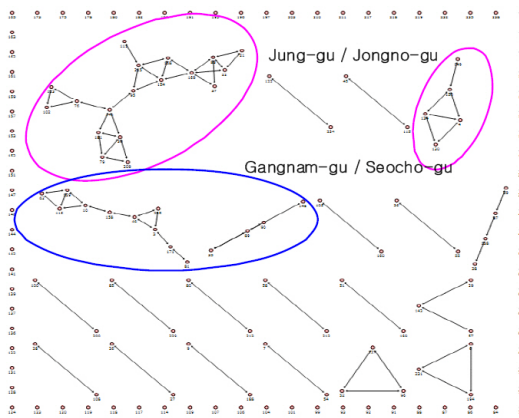


Figure 3. Starbucks network in Seoul

above, and analyzed the network influence models based on these Starbucks clusters.

4 Influence Models

4.1 Diffusion Mechanism

A store with large number of visitors is highly possible to become saturated with exponentially increasing visitors. To balance between supply and demand, it is possible (1) to open a new shop near the store, or (2) for visitors to move to neighbor stores looking for empty seats. This phenomenon would repeatedly occur in existing stores if demand exceeds supply. This is the basic idea of diffusion mechanism.

Based on this mechanism, we propose two network influence models, 'Dynamic Influence Model', and 'Static Influence Model' which will be explained in more detail in section 4.3.

In Dynamic Influence Model, there are two main factors to consider; *distance* and *time*. Both the distance between two stores and the time difference between the open-

ing years of them is important to decide an extent of influential power of each store for evaluating the temporal process of spreading.

However, Static Influence Model does not consider the factor of time. The reason is that it estimates the cumulative nodal influential power given the fixed size of network at current time.

4.2 Basic Assumptions

There are basic assumptions for evaluating the nodal diffusive power of Starbucks Network as follows.

- As the number of visitors of a store increases, additional neighbor stores are needed. (a balance of supply and demand)
- The influence score of store A decreases with the time difference of the opening dates from the neighbor store B. (except the case that the years of the opening dates are the same; This is because we assume that it takes almost one year to open a new shop. Furthermore, as the difference of the opening dates becomes larger, the influence power decreases non-linearly as Fig. 4.(Sigmoid Function)

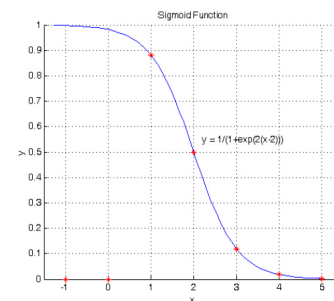


Figure 4. Sigmoid Function

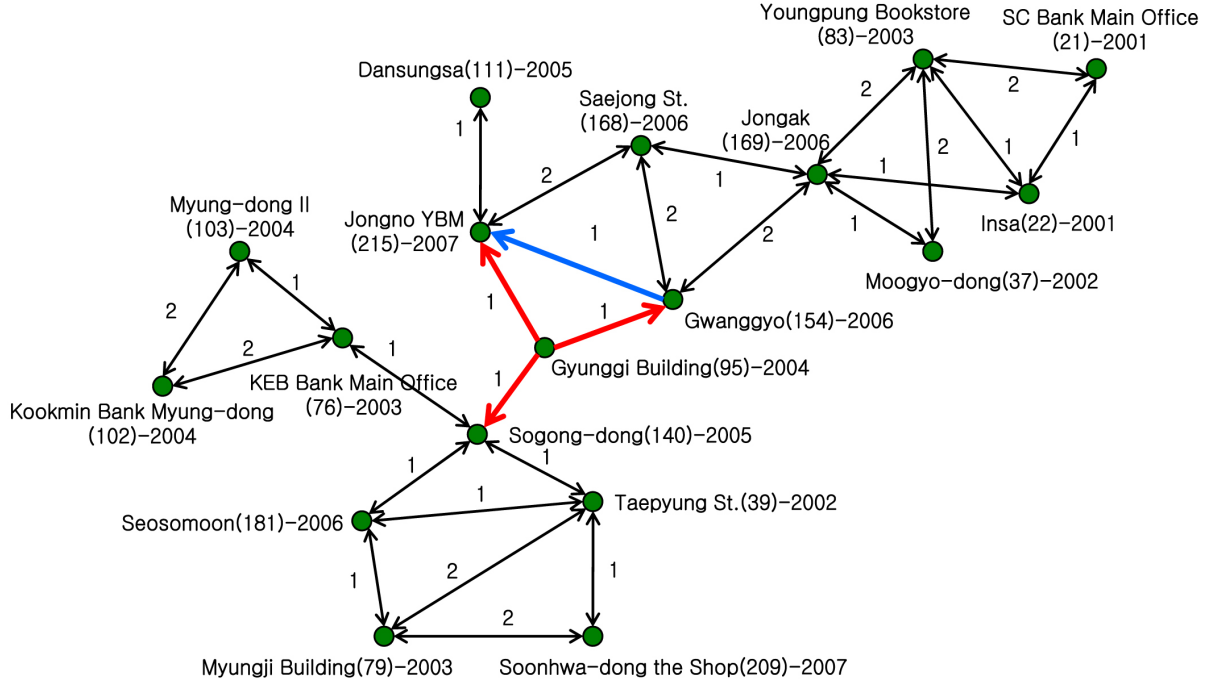


Figure 5. Starbucks Store Cluster of Jung-gu and Jongno-gu and its effect propagation procedures

4.3 Influence Models

Based upon the basic assumptions above, we defined the evaluation models for estimating the nodal influential power.

4.3.1 Dynamic Influence Model

We model the effect of store i , E_i as Fig. 6.

Here,

- $W(i, j)$: the weight of the distance between store i and store j
- f_{year} : the weight of the time difference between the opening years of two stores
- f_{step} : the damping factor as the propagation step goes further (here, we set it to 0.5)
- e : the initial influence score (here, we set it to 1.0)

The effect e of the store i is propagated to the connected neighbor store j , and the effect is propagated along to the next connected neighbor repeatedly. That is, the effect of a store is propagated recursively until there are no connected nodes any more and the opening year of store i is less than that of store j ; it is a directed graph. Every step, the effect will be recalculated in accordance with propagation weight.

$$E_i(e) = e + \sum_{j \in C_i} E_j \left(\underbrace{f_{step} \cdot f_{year}(i, j) \cdot W(i, j)}_{\text{propagated effect from node } i \text{ to node } j} \cdot e \right)$$

, where

$$C_i = \{j \mid W(i, j) > 0\} : \text{a set of all connected nodes,}$$

$$W(i, j) = \begin{cases} 1, & \text{if distance}(i, j) \in [0, 200\text{m}) \\ 1/2, & \text{if distance}(i, j) \in [200\text{m}, 400\text{m}) \\ 1/3, & \text{if distance}(i, j) \in [400\text{m}, 600\text{m}) \\ 0, & \text{otherwise} \end{cases}$$

$$f_{year}(i, j) = \begin{cases} \frac{1}{1 + e^{2(\text{year}(j) - \text{year}(i) - 2)}}, & \text{if } \text{year}(i) < \text{year}(j) \\ 0, & \text{otherwise} \end{cases}$$

$$f_{step} = 0.5$$

$\therefore$ Effect of node $i \equiv E_i(1)$

Figure 6. Dynamic Influence Model

Fig. 5 shows the process of the effect propagation. The red arrows mean the 1st step of the effect propagation from a seed node i and the blue arrows indicate the 2nd step of the effect propagation from the end node of the 1st step.

As we mentioned above, it is a directed graph; the head of an arrow indicates the target node, store j which is opened

Table 1. Results from Dynamic Influence Model and Static Influence Model for Jung-gu/Jongno-gu Starbucks cluster

Dynamic Influence Model					Static Influence Model				
Rank	Store Name	Open Order	Open Year	Influence Score	Rank	Store Name	Open Order	Open Year	Influence Score
1	Gyunggi Building	95	2004	2.054	1	Sogong-dong	140	2005	14.810
2	KEB Main Office	76	2003	2.021	2	Jonggak	169	2006	14.370
3	Gwanggyo	154	2006	1.440	3	Jongno YBM	215	2007	13.679
3	Sogong-dong	140	2005	1.440	4	Gwanggyo	154	2006	12.049
5	Taepyung St.	39	2002	1.330	5	Taepyung St.	39	2002	10.759
6	Insa	22	2001	1.259	6	Youngpung Bookstore	83	2003	10.496
7	Dansungsa	111	2005	1.250	7	Gyunggi Building	95	2004	9.262
8	Moogyo-dong	37	2002	1.236	8	Seosomoon	181	2006	9.170
9	Saejong St.	168	2006	1.220	9	KEB Main Office	76	2003	8.131
10	SC Bank Main Office	21	2001	1.129	10	Saejong St.	168	2006	7.881
11	Myunggi Building	79	2003	1.064	11	Insa	22	2001	7.817
12	Youngpung Bookstore	83	2003	1.030	12	Myunggi Building	79	2003	7.661
13	Jongno YBM	215	2007	1.000	13	Myung-dong II	103	2004	6.791
13	Jonggak	169	2006	1.000	14	Kookmin Bank Myung-dong	102	2004	6.541
13	Myung-dong II	103	2004	1.000	15	Soonhwa-dong the Shop	209	2007	5.539
13	Kookmin Bank Myung-dong	102	2004	1.000	16	SC Bank Main Office	21	2001	5.327
13	Seosomoon	181	2006	1.000	17	Moogyo-dong	37	2002	4.810
13	Soonhwa-dong the Shop	209	2007	1.000	18	Dansungsa	111	2005	2.500

later than the source node, store i .

The link weight represents the distance between the two stores (1: ~200m, 2: 200m~400m, 3: 400m~600m) and the node labels indicate ‘the name of stores (the opening order)-the year of opening date’.

$$E_i(s) = \begin{cases} CE_i(s) & , \text{if } CE_i(s) < Threshold_{effect} \\ CE_i(s) + \sum_{j \in C_i} E_j(s+1), & \text{otherwise} \end{cases}$$

step effect of current step effect of next step

, where

$$CE_i(s) = f_{step}(s) \sum_{j \in C_i} W(i, j) : \text{current effects on connected nodes}$$

$$C_i = \{j \mid W(i, j) > 0\} : \text{a set of all connected nodes,}$$

$$W(i, j) = \begin{cases} 1, & \text{if distance}(i, j) \in [0, 200\text{m}) \\ 1/2, & \text{if distance}(i, j) \in [200\text{m}, 400\text{m}) \\ 1/3, & \text{if distance}(i, j) \in [400\text{m}, 600\text{m}) \\ 0, & \text{otherwise} \end{cases}$$

$$f_{step}(s) = 0.5^s$$

$$Threshold_{effect} = 0.0001$$

$\therefore$ Effect of node $i \equiv E_i(1)$

Figure 7. Static Influence Model

4.3.2 Static Influence Model

Static Influence Model is to evaluate the nodal influence power at the current point, so the size of the network is not dynamic but fixed. While the previous model, Dynamic Influence model considers the time factor for estimating an extent of contribution of each node to expand the network size, this model excludes it. However, the basic mechanism is almost the same. Also, in this model, the propagation process is recursively continued until the value of effect converges into a constant value; it is an undirected graph. Following equation describes the logic mentioned above.

5 Simulation Results

Table 1 shows the results from Dynamic Influence Model and Static Influence Model for Jung-gu/Jongno-gu Starbucks cluster respectively. As you can see, the results from the two models are different; by the Dynamic Influence Model, the most influential store is 95th shop, named Gyunggi-Building, while by the Static Influence Model, the most influential store is 140th shop, named Sogong-dong. You can see the picture of the 1st influential store, Gyunggi-Building by the Dynamic Influence Model in Fig. 8 and its surroundings in Fig. 9.

As you can see Fig. 8, the Scale of the Gyunggi-Building store is relatively small to other Starbucks stores in Korea but the surroundings of the store in Fig. 9 are very advantageous for a floating population to be gathered due to a lot of convenience facilities such as a bank, a large-scale



Figure 8. Gyunggi-Building Starbucks Store (The most influential store in expanding the size of Starbucks Network in Jung-gu/Jongno-gu)

book store, subway station, department store, office buildings, and so on. Moreover, the office workers with Starbucks Coffee in their hands play a role of advertising the Starbucks brand to other people on the street.

Therefore, it is reasonable that Gyunggi-Building store is the most influential in expanding the size of Starbucks Network in Jung-gu/Jongno-gu districts. That is, Starbucks Network Analysis can help to find out the powerful locations to earn the word of mouth effect most efficiently.

However, due to the increase of other coffee brands, the Starbucks Network is not independent any more. Instead, it combines other coffee brands' network such as The Coffee Bean and Tea Leaf, Dunkin Donuts, and so on. This phenomenon brings the changes of influential power as you can see Fig. 10 and Table 2.

As the Table 2 shows, even though the number of Coffee Bean stores is relatively small to that of Starbucks stores, a large portion of Coffee Bean stores are ranked highly in influence score lists above. Moreover, the most influential store is Coffee Bean, not Starbucks. However, nodes in a network are affected by each other, so the rank can be always changed by the position of a new store.

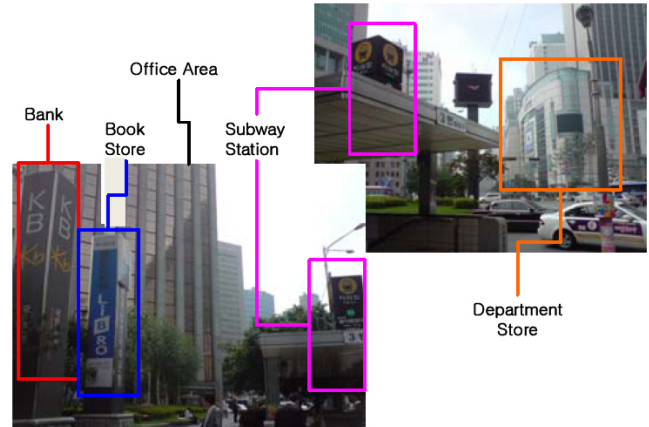


Figure 9. The Surroundings of Gyunggi-Building Starbucks Store

Table 2. Influence Scores by Static Influence Model for Jung-gu/Jongno-gu Starbucks-CoffeeBean Network

Rank	Brand	Store Name
1	Coffee Bean	Sogong-dong Sanhwa Building
2	Starbucks	KEB Main Office
3	Coffee Bean	Myung-dong
4	Coffee Bean	Myung-dong Securities
5	Starbucks	Myung-dong II
6	Starbucks	Kookmin Bank Myung-dong
7	Starbucks	Sogong-dong
8	Starbucks	Taepyung St.
9	Starbucks	Gyunggi Building
10	Starbucks	Jongno YBM
11	Coffee Bean	Jongno Gwanchul-dong
12	Starbucks	Gwanggyo
13	Starbucks	Seosomoon
14	Starbucks	Saejong St.
15	Coffee Bean	Jongno Cinecore
16	Starbucks	Myungji Building
17	Coffee Bean	Taepyung St.
18	Starbucks	Jongak
19	Starbucks	Soonhwa-dong the Shop
20	Starbucks	Youngpung Bookstore

6 Conclusion

The success of Starbucks Korea is due to the diffusive power of network which is formed by the word-of-mouth effect based on the customer social network. In addition, to obtain the word-of-mouth effect most efficiently, you should consider the surroundings of a store like Fig. 10. Also, you would better to open a new shop near the powerful store to keep the chain reaction. If the Starbucks stores are isolated or are not located in powerful areas, Starbucks

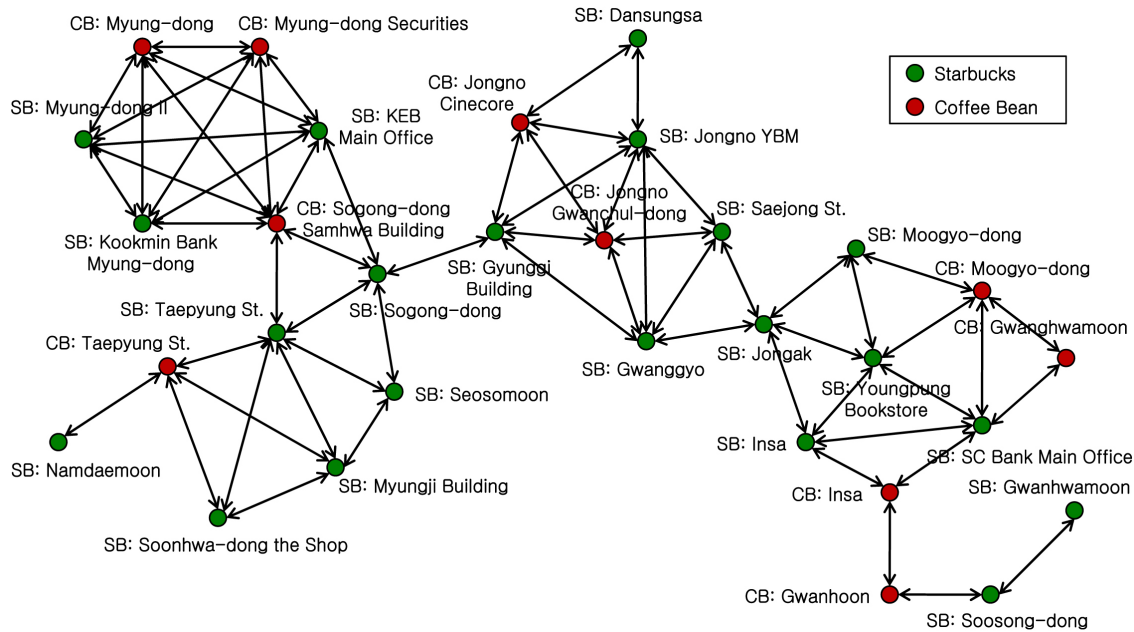


Figure 10. Starbucks-CoffeeBean Store Cluster of Jung-gu and Jongno-gu

Korea might not achieve today's outcome.

In addition to the internal correlation of a social network, you should remember the interaction to the surroundings. That is why we considered Coffee Bean stores in this paper. Therefore, not only opening a new shop in powerful area but also considering positions in a whole network including every coffee brands is important.

There were research limitations in this paper. If we could obtain the information about the number of visitors of each store, the type of visitors (age, sex, occupation, etc), the profit of each store, etc, we could develop more concrete models.

The proposed models can be applied to other industry for evaluating current distribution of shops or for making a decision of location to launch a new brand shop. In addition, we expect these models can contribute to various kinds of domains for evaluating the location based nodal influential power.

References

- [1] *The Coffee Bean and Tea Leaf Korea homepage* (www.coffeebeankorea.com).
- [2] *Starbucks Korea homepage* (www.istarbucks.co.kr).
- [3] P. Carrington, J. Scott, and S. Wasserman. *Models and methods in social network analysis*. Cambridge University Press, 2005.
- [4] W. De Nooy, A. Mrvar, and V. Batagelj. *Exploratory social network analysis with Pajek*. Cambridge University Press New York, NY, USA, 2004.
- [5] P. Domingos. Mining social networks for viral marketing. *IEEE Intelligent Systems*, 20(1):80–82, 2005.
- [6] J. Goldenberg, B. Libai, and E. Muller. Talk of the network: A complex systems look at the underlying process of word-of-mouth. *Marketing Letters*, 12(3):211–223, 2001.
- [7] D. Kempe, J. Kleinberg, and É. Tardos. Influential nodes in a diffusion model for social networks. *Lecture notes in computer science*, pages 1127–1138.
- [8] D. Kempe, J. Kleinberg, and É. Tardos. Maximizing the spread of influence through a social network. In *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 137–146. ACM New York, NY, USA, 2003.
- [9] V. Mahajan, E. Muller, and F. Bass. New product diffusion models in marketing: A review and directions for research. *The Journal of Marketing*, pages 1–26, 1990.
- [10] M. Meng. *The secret of Starbucks Korea's 100th Store*. Vision Korea, 2005.
- [11] E. Rogers. *Diffusion of innovations*. Free Press, 1995.
- [12] T. Schelling. *Micromotives and macrobehavior*. 1978.
- [13] S. Wasserman and K. Faust. *Social network analysis: Methods and applications*. Cambridge Univ Pr, 1994.
- [14] D. Watts. A simple model of global cascades on random networks. *Proceedings of the National Academy of Sciences*, 99(9):5766, 2002.
- [15] H. Young. The diffusion of innovations in social networks. *The Economy As an Evolving Complex System III: Current Perspectives and Future Directions*, page 267, 2006.